Student Performance Dashboard Recreation

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1. Introduction

This report analyzes the Students Performance dataset obtained from Kaggle.

Dataset Source:
Students Performance in Exams (Kaggle Dataset)

The objective is to perform Exploratory Data Analysis (EDA) and create visualizations using RStudio.

2. Loading Required Libraries

3. Import Dataset

```r
data <- read.csv("StudentsPerformance.csv")
head(data)
```

```
##   gender race.ethnicity parental.level.of.education        lunch
## 1 female        group B           bachelor's degree     standard
## 2 female        group C                some college     standard
## 3 female        group B             master's degree     standard
## 4   male        group A          associate's degree free/reduced
## 5   male        group C                some college     standard
## 6 female        group B          associate's degree     standard
##   test.preparation.course math.score reading.score writing.score
## 1                    none         72            72            74
## 2               completed         69            90            88
## 3                    none         90            95            93
## 4                    none         47            57            44
## 5                    none         76            78            75
## 6                    none         71            83            78
```

```r
str(data)
```

```
## 'data.frame':    1000 obs. of  8 variables:
##  $ gender                     : chr  "female" "female" "female" "male" ...
##  $ race.ethnicity             : chr  "group B" "group C" "group B" "group A" ...
##  $ parental.level.of.education: chr  "bachelor's degree" "some college" "master's degree" "associate
```

```
##  $ lunch                      : chr  "standard" "standard" "standard" "free/reduced" ...
##  $ test.preparation.course    : chr  "none" "completed" "none" "none" ...
##  $ math.score                 : int  72 69 90 47 76 71 88 40 64 38 ...
##  $ reading.score              : int  72 90 95 57 78 83 95 43 64 60 ...
##  $ writing.score              : int  74 88 93 44 75 78 92 39 67 50 ...
```

```r
summary(data)
```

```
##     gender          race.ethnicity     parental.level.of.education
##  Length:1000        Length:1000        Length:1000
##  Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character
##
##
##
##     lunch           test.preparation.course   math.score     reading.score
##  Length:1000        Length:1000             Min.   :  0.00   Min.   : 17.00
##  Class :character   Class :character        1st Qu.: 57.00   1st Qu.: 59.00
##  Mode  :character   Mode  :character        Median : 66.00   Median : 70.00
##                                             Mean   : 66.09   Mean   : 69.17
##                                             3rd Qu.: 77.00   3rd Qu.: 79.00
##                                             Max.   :100.00   Max.   :100.00
##  writing.score
##  Min.   : 10.00
##  1st Qu.: 57.75
##  Median : 69.00
##  Mean   : 68.05
##  3rd Qu.: 79.00
##  Max.   :100.00
```

4. Data Cleaning

```r
colSums(is.na(data))
```

```
##                      gender              race.ethnicity
##                           0                           0
## parental.level.of.education                       lunch
##                           0                           0
##     test.preparation.course                  math.score
##                           0                           0
##               reading.score               writing.score
##                           0                           0
```

```r
data$gender <- as.factor(data$gender)
data$race.ethnicity <- as.factor(data$race.ethnicity)
data$parental.level.of.education <- as.factor(data$parental.level.of.education)
data$lunch <- as.factor(data$lunch)
data$test.preparation.course <- as.factor(data$test.preparation.course)
```

5. Feature Engineering

```
data$total_score <- data$math.score +
                    data$reading.score +
                    data$writing.score

data$average_score <- data$total_score / 3

summary(data$average_score)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    9.00   58.33   68.33   67.77   77.67  100.00
```

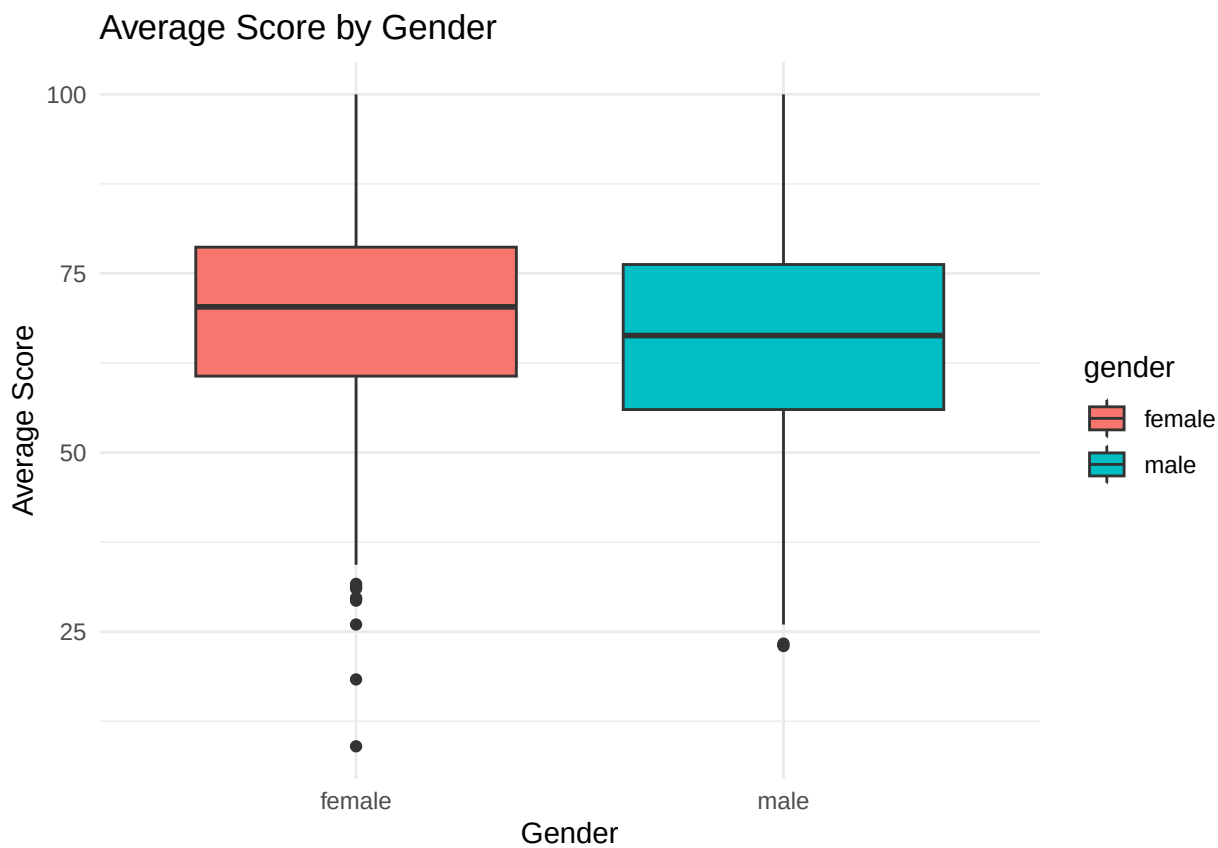
6. Gender-wise Performance

```
gender_avg <- data %>%
  group_by(gender) %>%
  summarise(
    Math = mean(math.score),
    Reading = mean(reading.score),
    Writing = mean(writing.score),
    Average = mean(average_score)
  )

gender_avg
```

```
## # A tibble: 2 x 5
##   gender  Math Reading Writing Average
##   <fct>  <dbl>   <dbl>   <dbl>   <dbl>
## 1 female  63.6    72.6    72.5    69.6
## 2 male    68.7    65.5    63.3    65.8
```

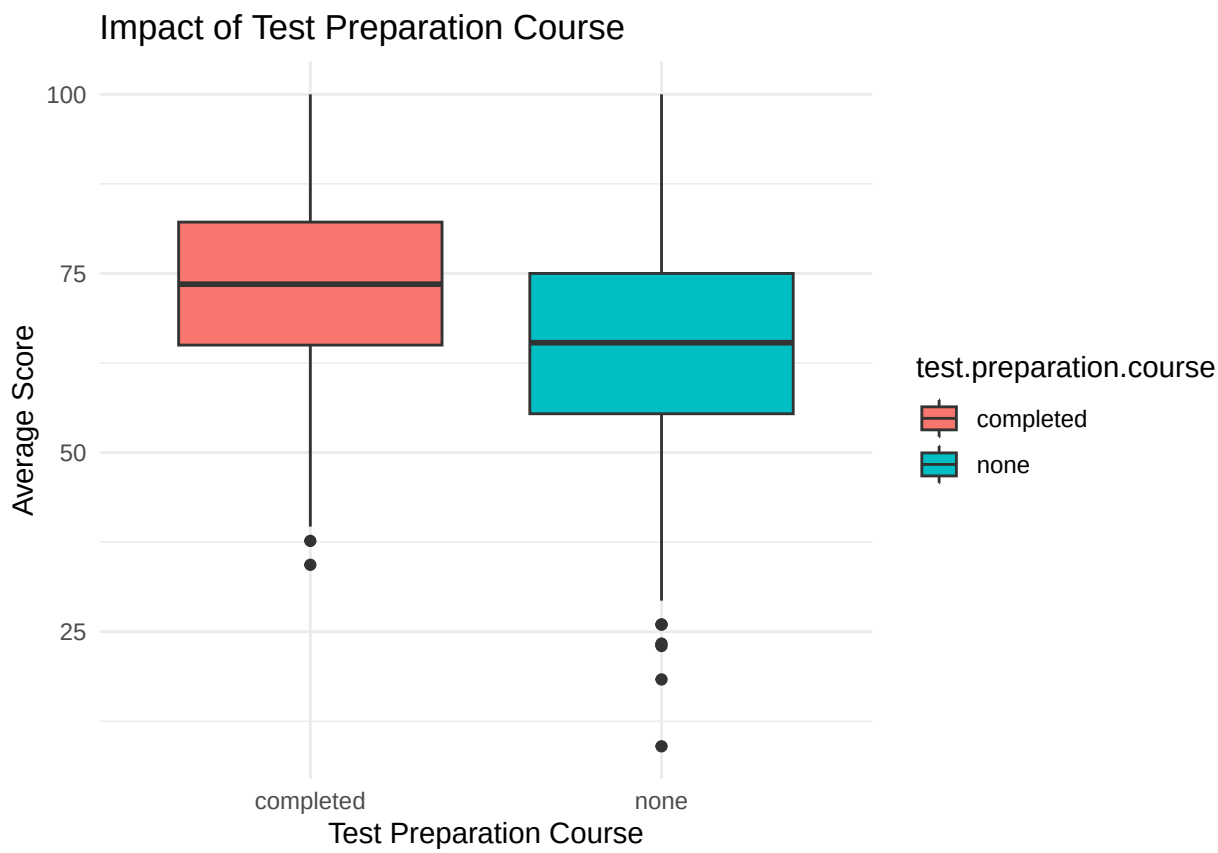
```
ggplot(data, aes(x = gender, y = average_score, fill = gender)) +
  geom_boxplot() +
  theme_minimal() +
  labs(title = "Average Score by Gender",
       x = "Gender",
       y = "Average Score")
```



Observation:
Female students perform better in reading and writing, while math performance is comparable.

7. Test Preparation Impact

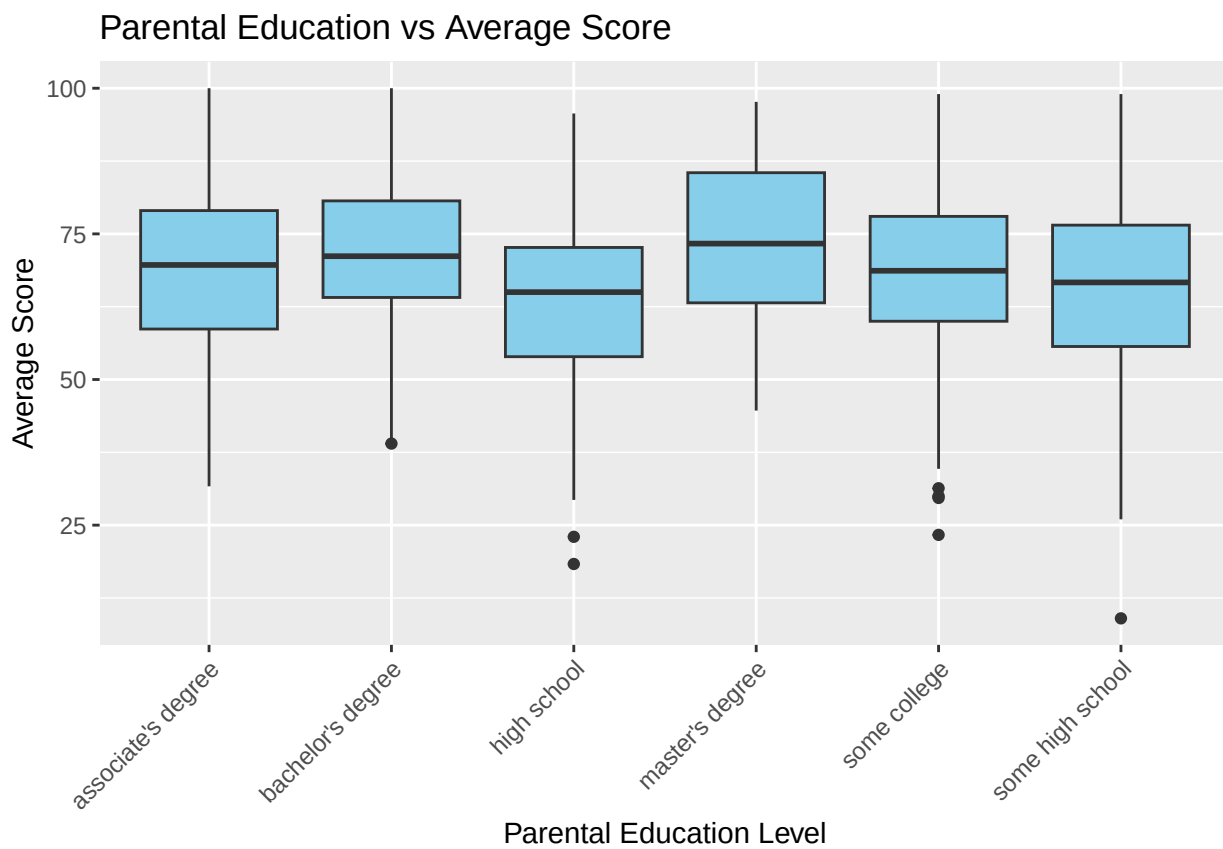
```
ggplot(data, aes(x = test.preparation.course,
                 y = average_score,
                 fill = test.preparation.course)) +
  geom_boxplot() +
  theme_minimal() +
  labs(title = "Impact of Test Preparation Course",
       x = "Test Preparation Course",
       y = "Average Score")
```



Observation:
Students who completed the test preparation course have higher average scores.

8. Parental Education Impact

```
ggplot(data, aes(x = parental.level.of.education,
                 y = average_score)) +
  geom_boxplot(fill = "skyblue") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Parental Education vs Average Score",
       x = "Parental Education Level",
       y = "Average Score")
```



Observation:
Higher parental education levels are associated with better student performance.

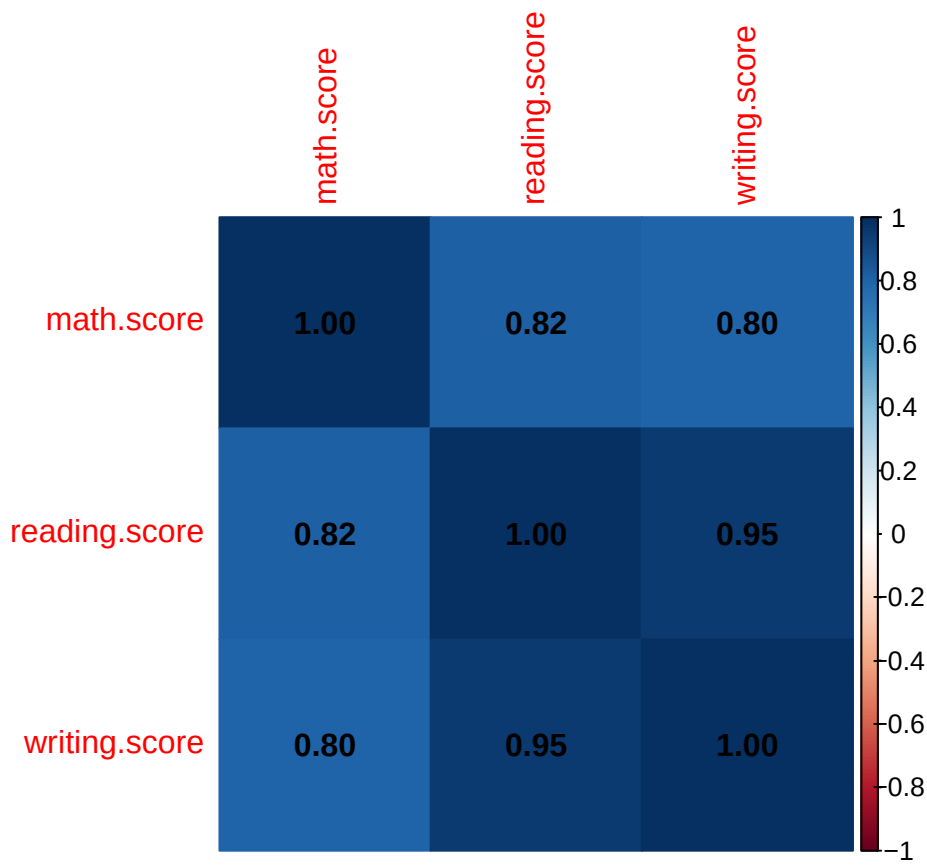
9. Correlation Analysis

```
score_data <- data[, c("math.score",
                       "reading.score",
                       "writing.score")]

cor_matrix <- cor(score_data)
cor_matrix
```

```
##               math.score reading.score writing.score
## math.score     1.0000000     0.8175797     0.8026420
## reading.score  0.8175797     1.0000000     0.9545981
## writing.score  0.8026420     0.9545981     1.0000000
```

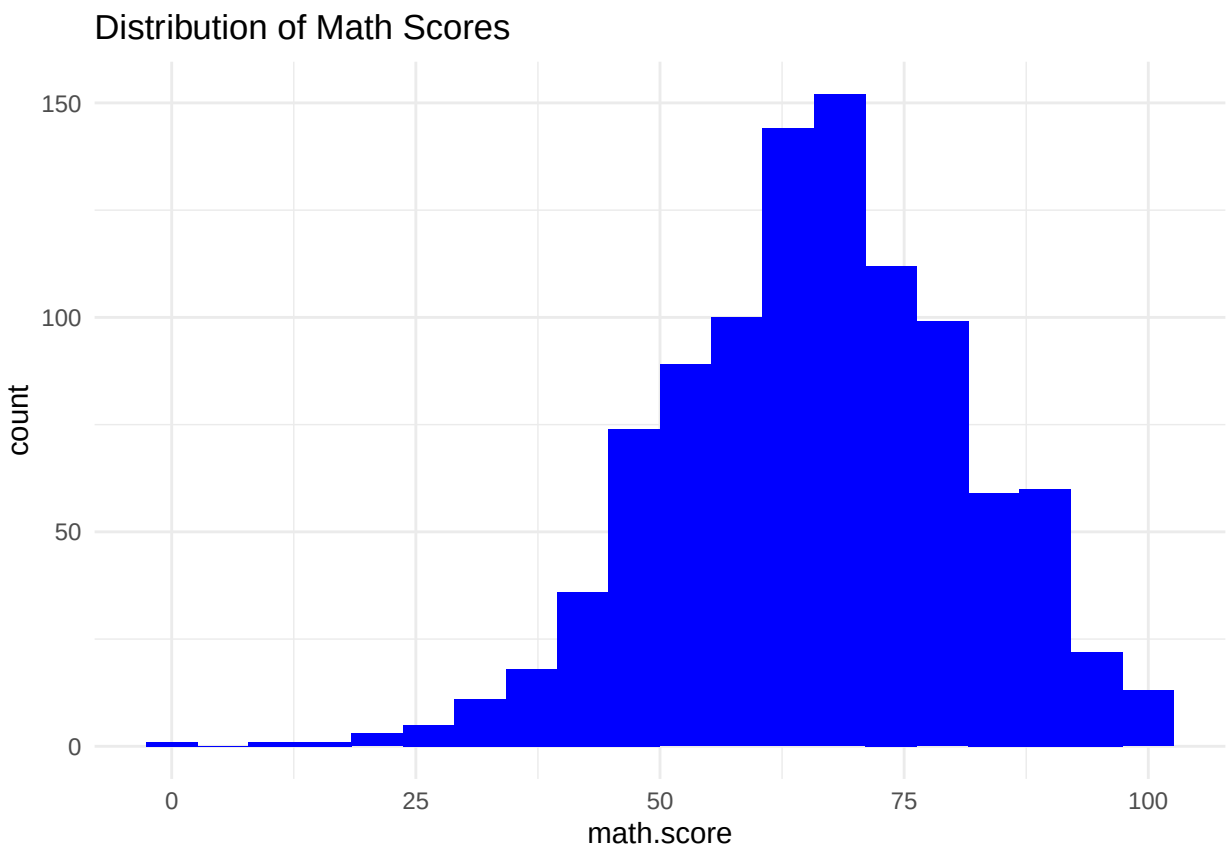
```
corrplot(cor_matrix,
         method = "color",
         addCoef.col = "black")
```



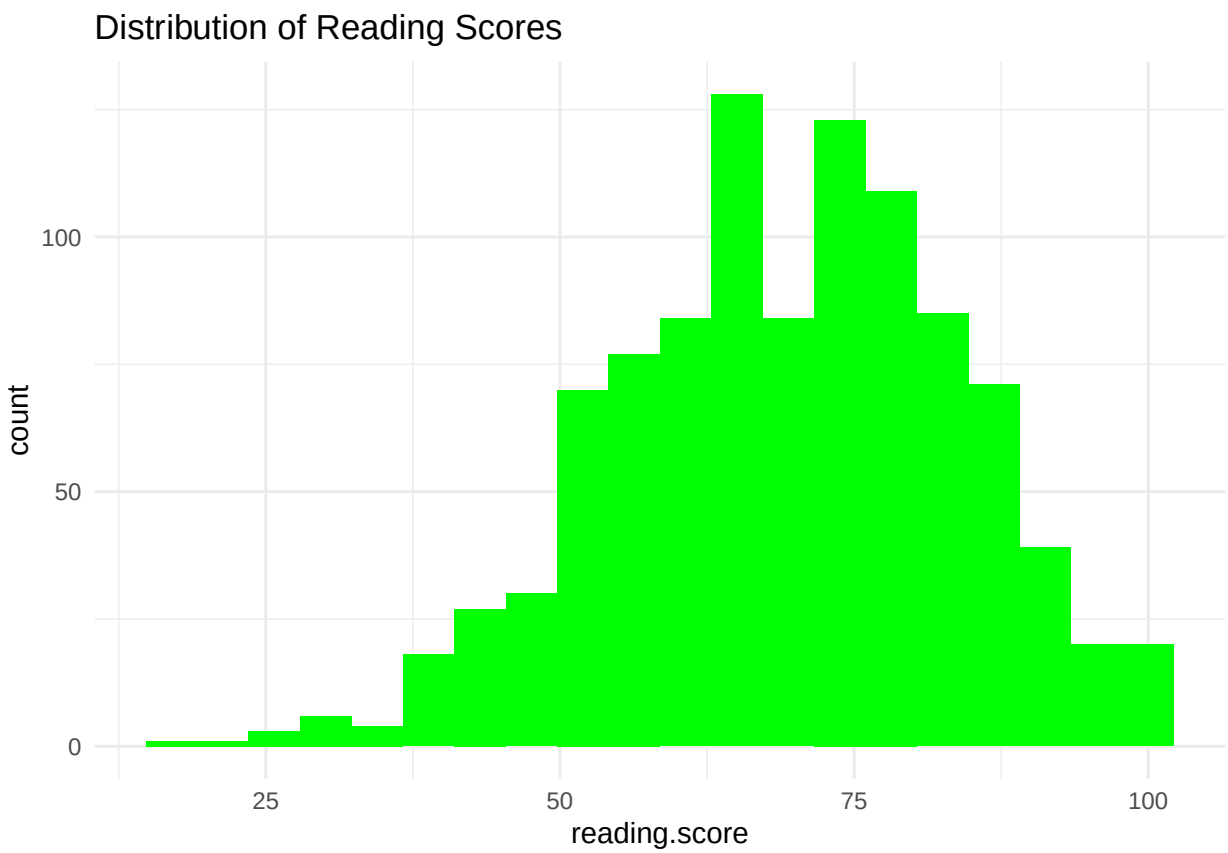
Observation:
There is a strong positive correlation between all three subjects.

10. Score Distribution

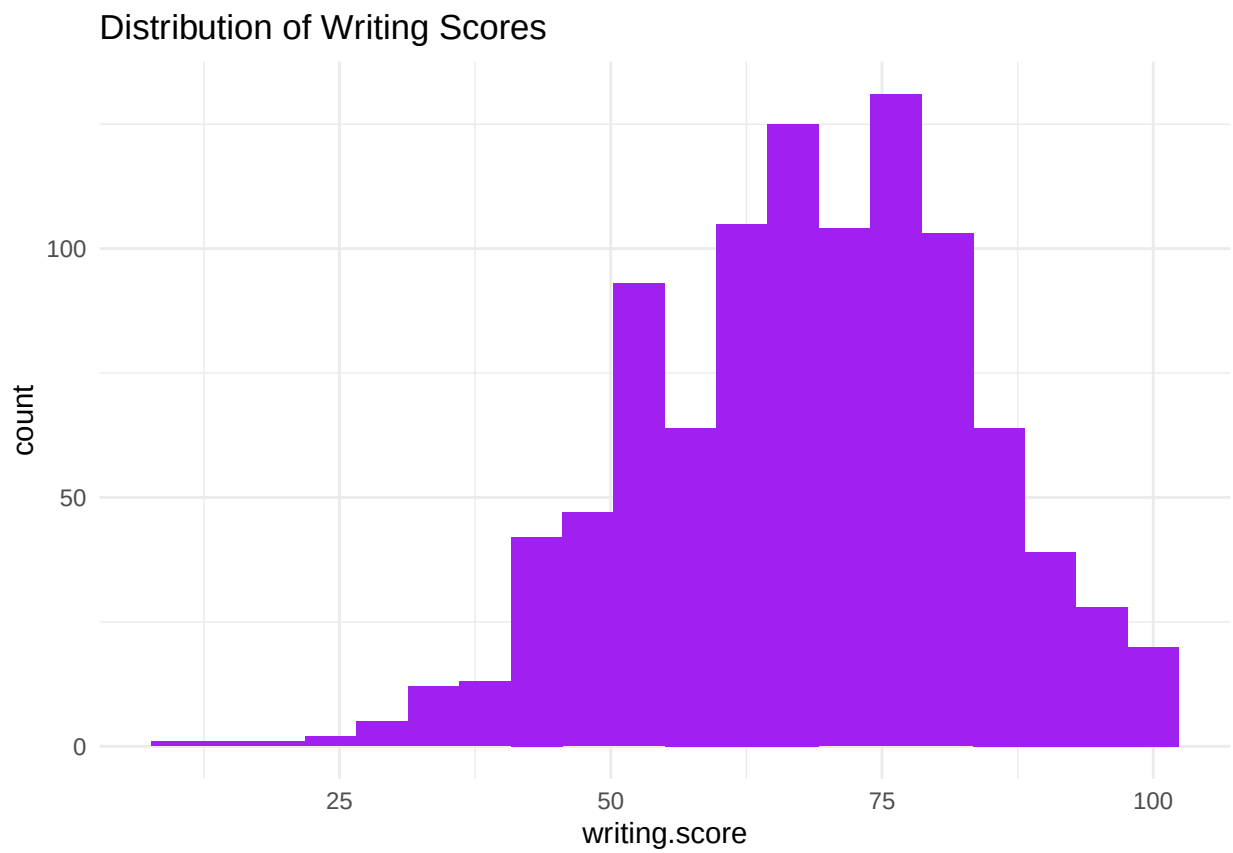
```
ggplot(data, aes(x = math.score)) +
  geom_histogram(fill = "blue", bins = 20) +
  theme_minimal() +
  labs(title = "Distribution of Math Scores")
```




```r
ggplot(data, aes(x = reading.score)) +
  geom_histogram(fill = "green", bins = 20) +
  theme_minimal() +
  labs(title = "Distribution of Reading Scores")
```




```r
ggplot(data, aes(x = writing.score)) +
  geom_histogram(fill = "purple", bins = 20) +
  theme_minimal() +
  labs(title = "Distribution of Writing Scores")
```



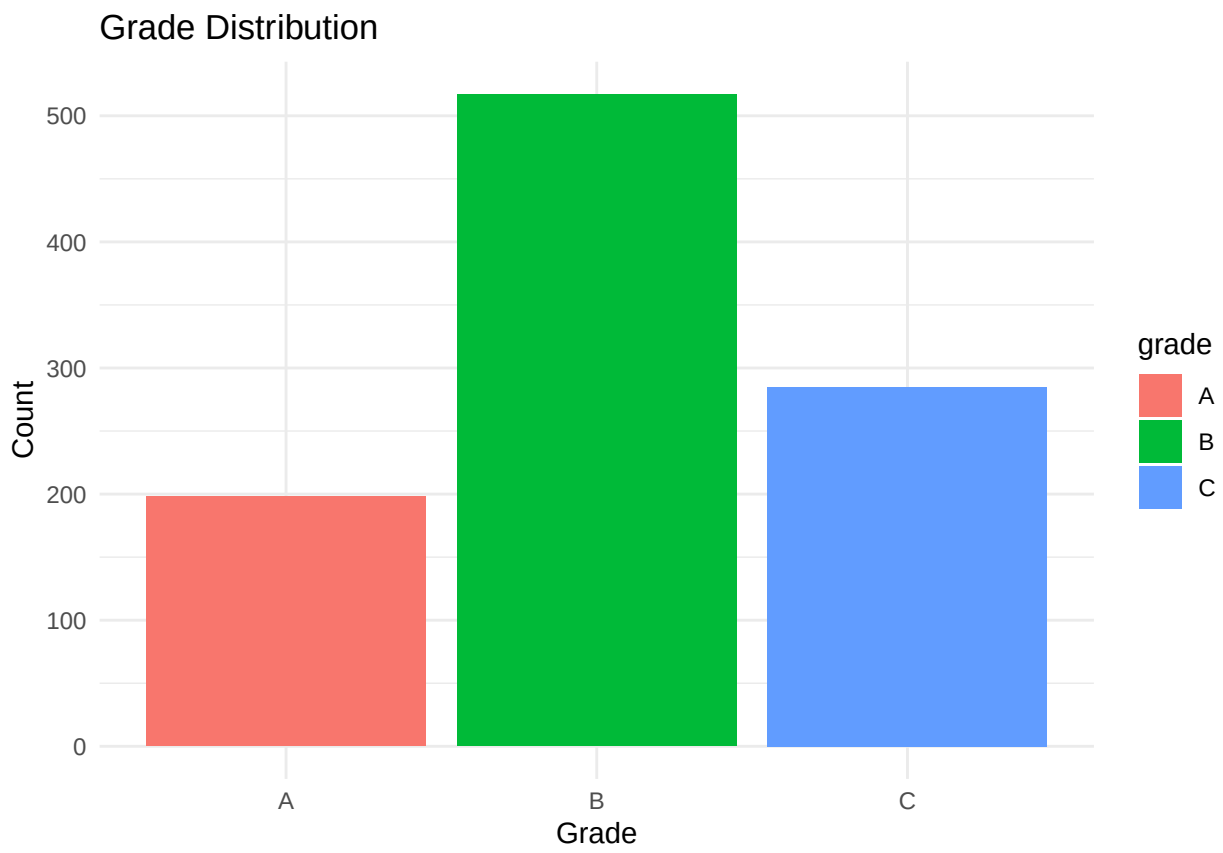
11. Grade Classification

```
data$grade <- ifelse(data$average_score >= 80, "A",
                     ifelse(data$average_score >= 60, "B", "C"))

table(data$grade)
```

```
##
##   A   B   C
## 198 517 285
```

```
ggplot(data, aes(x = grade, fill = grade)) +
  geom_bar() +
  theme_minimal() +
  labs(title = "Grade Distribution",
       x = "Grade",
       y = "Count")
```



Observation:
Most students fall into Grade B category.

12. Conclusion

- Test preparation improves student performance.
- Strong positive correlation exists among math, reading, and writing scores.
- Parental education positively influences student outcomes.
- Female students perform better in reading and writing subjects.